

Method for Mobile Robot Navigation Using Reinforcement Learning in an Unfamiliar and Complicated Environment

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Abstract— In recent years, mobile robots (MRs) have seen explosive growth in use across several industries, including production, healthcare, security, as well as the automation of warehouses. Ensuring the safety and effectiveness of these MRs' functioning requires the design of efficient control mechanisms that can adapt to changing situations. Give a new way to manage MRs using reinforcement learning (RL) in this paper. The process begins with the development of a mathematical model, followed by the training of a neural network (NN) to create a control plan for the robot using reinforcement learning (RL). The policy is learned by MR via trial and error as it explores its environment and is rewarded based on its behavior. The robot will be more motivated to avoid obstacles and get toward its aim if it is offered incentives. Hiring a deep Q-learning (QL) agent has opened up new possibilities for robot navigation, including the ability to learn successfully traverse unfamiliar terrains while avoiding hazards. After an RL model finds the target location using MR autonomously in an unknown territory, a Deep Q-Network (DQN) is used to get there. Evaluation of our methods via the usage of an Epsilon-Greedy algorithm simulation. In comparison to existing MR monitoring systems, the technique is more efficient and safer, according to the findings.

Keywords— Navigation; Mobile Robot; Reinforcement Learning; Q-Network; Algorithm

I. INTRODUCTION

The rapid development of AI (artificial technology) is enabling MRs enhance their capabilities in doing things which used to be considered too complex and challenging, including heavy-weight transportation, space exploration, security, and various other fields [1]. When artificial intelligence and robots are combined, robots have the potential to become more intelligent and capable of doing jobs independently of humans. MRs are becoming increasingly effective because to AI technology; they have potential applications in many sectors, including as healthcare, security, postal delivery, infrastructure inspection and construction, the military, and freight [2]. Aside from other technologies, MR uses the sensors to determine its surroundings and complete its mission [3]. Determining the most efficient routes to the target while avoiding obstacles is a crucial component of MRs navigation process. The most important part of a navigation system is figuring out the distance something may advance on a certain amount of power [3], [5], [6], [7]. Due to its reliability in generating results, intelligence (AI) methods are being used

by several studies in order to make the path creation procedure more automated. Among the trickiest parts of making robots is figuring out their paths [8]. Concerning the development of MRs for practical usage, experts believe that the most important factor describes work across the tropic's climates. The standard MR navigational technique lacks the potential to enable learning on its own [9], [10]. MRs have the capability to independently traverse uncharted territories while circumventing both fixed and dynamic impediments. Numerous real-world issues, most notably MR navigation, make use of RL ideas, and the RL approach achieved tremendous progress [4]. There has been a lot of buzz about RL's potential to solve the basic navigation problem that MRs face because of its flexible over experiences-based Q-learning and good visuals [11]. Robots on wheels need to be able to navigate congested areas with ease and communicate with humans and other obstacles without causing any harm [12]. Another obstacle to learning-based collision avoidance is that deep learning in robotics does not guarantee the correctness or dependability of a learnt policy. Due to the difficulty in producing secure and efficient instructional information over physical robots or the fact that the quantity and quality of information obtained greatly impact on the efficacy of robot learning, the collision-avoidance rules utilized within the present investigation has been taught in a simulated environment instead of the actual world [13], [14]. Figure 1 shows that the traditional approach to navigation uses simultaneous localization and mapping (SLAM) to map an unfamiliar area, an orientation localization method to find where the MR is presently, and a route planning mechanism to guide it to its destination. Conventional navigational frameworks fail in novel and unpredictable environments because they rely on very accurate world visualization, that's susceptible to sensory interference. Given that computation stress grows dramatically with increasing input variety, the specified complexity of multimodal scenarios poses a problem for the conventional DRL technique [4], [15]. Training a deep neural network (DNN) yields nonlinear models; this is how the deep learning (DL) approach learns the characteristics and rules of the initial input. Deep reinforcement learning (DRL) is the modern name for a combination of RL and a DNN [16]. In order to supplement or perhaps replace the traditional method of navigation, many research are now using DRL-based MR navigation [17].

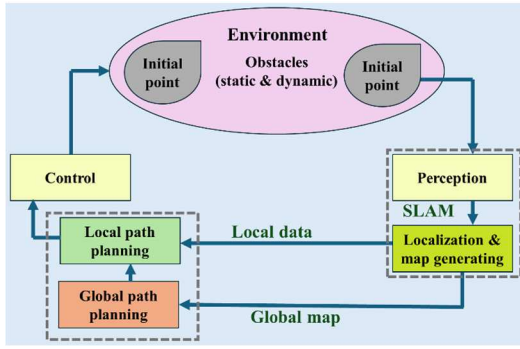


Fig. 1. Showing the conventional method of MR navigation

Navigators from all around the world have been attempting to solve the problem for the last 20 years. The use of a mix of methods is a popular tactic. Recent years have seen a proliferation of deep learning and particularly DRL-based methodologies—that have made map-free navigation a reality, thanks to deep neural networks' exceptional modelling capabilities and RL's strong interaction, exploration, and self-learning capabilities [4]. A novel method for DRL training is presented in, which drastically reduces training duration. This study enhances the visual navigational capabilities of robots fitted with cameras [19]. The navigation problems may be categorized into many manageable fundamental tasks via a decentralized learning methodology that incorporates Deep Reinforcement Learning (DRL) [20]. This will ensure that the robot can navigate a big and complicated interior region without a map.

This paper presents DDRL-based decision strategy that utilizes a deep neural network to train a research method while considering an incomplete mapping. The authors Cai et al. [22] provide a deep policy gradient method based on Linear Temporal Logic (LTL) formulas for controlling a dynamically uncertain robot in a congested workplace. Using sampling-based approaches, this research introduces a novel method for guiding routes and rewarding users to address longevity problems throughout the instruction process and effectively completes target-reaching tasks. To facilitate safe and efficient navigation, Yan et al. [23] propose a multirobot navigation architecture based on option-based hierarchy DDRL. A lower-level control scheme uses sub-policies to generate actions in this architecture, whereas a higher-level control scheme produces a stable and dependable behavior selection policy on its own. In order to facilitate autonomous exploration of novel surroundings, Ali et al. provide a method to use DDRL. External, curiosity-driven reward systems are used to investigate the surroundings in this research. The outside monetary incentive assists the participant in preventing accidents, while the curiosity-driven incentive structure encourages the creature to explore uncharted territories by predicting future states. The primary objective of Xue et al. [24] was to develop DDRL algorithms that could efficiently navigate unmanned aerial vehicles (UAVs) over large-scale, unfamiliar areas with many obstacles. This study modelled an UAV using perception-constrained navigating using the partially observable Markov decision process (MDP) and then set up an actor critic orientated rapid recurring probabilistic valued gradients method for a real-time solution. Implementing a hybrid approach that combines model-based control, Wang et al. provide a hierarchical navigation system [25]. In order to train a DDRL network to prevent collisions, one may use the SAC (Soft Actor-Critic) and LSTM (Long Short-Term Memory) methods for

predicting two-dimensional laser inputs, target positions, and robot states into a collection of local checkpoints. A novel DDRL method developed by Zeng et al. allows non-holonomic robots to be precisely controlled while they navigate through an unknown and unpredictable environment with obstacles in motion [26]. Results from simulations show that the proposed method outperforms conventional methods of robotic navigation in challenging and unfamiliar environments by continually giving acceleration commands. With an outstanding sampling rate and simulation-to-real transferring capability, Zhu et al. provide a hierarchical DRL architecture for safe and fast navigation: The minimum DRL policy allows the robot to follow the destination point while keeping a safe distance from obstacles [27]. To further improve navigational dependability, a high degree-based DRL stance provides applied. In order to minimize state space and avoid under rewarding the robot, this strategy selects a waypoint on a trajectory that takes it to the target location. In addition, the route is built using a global or regional map, which might significantly improve the recommended flexibility, safety, and accuracy of the DRL framework's samples.

II. METHODOLOGY

Acquiring sensor data and concurrently creating an environment map allows MR to execute autonomous navigation while traveling to a new region. For a more realistic-looking map, the MR exploration method is necessary. The MR often detects its own position and builds an environment map independently in traditional navigation methods. Many end-to-end control strategies have been developed with the help of DL and RL as they have evolved over the years. These innovations enable MR control by feeding data from sensors combined with the natural surroundings to DNN. Conciseness is a big plus for end-to-end methods. Regardless, a considerable amount of time must pass before the procedure can be used to a real MR. Many researchers have used end-to-end learning in simulation settings because of the high degree of trial and error involved [28]. The technique takes some time to converge as the MR starts by learning to navigate in order to explore new environments. There is no guarantee that DNN's learned mapping will be more applicable outside of a simulation situation because to the inherent differences between the two. It is possible that the MDP represents the connection between agents and their environment. In order to determine the best course of action for the framework, MDPs take a number of factors into account, including the environment, agent behavior, and incentives. A Markov decision process (MDP) is a description of an iterative decision-making issue where an agent has to choose from a set of options in order to maximize a metric based on rewards [29]. An MDP (M) could be abbreviated as Equation (1).

$$M = (S, A, T, R) \quad (1)$$

The given sentence describes the relationship between S, the discrete set of possible states A, the range of potential steps to take $T(s, s') = \text{Tr}(st+1 = s' | st = s, at = a)$, Which of the possibility that the operation is taken for states s at the moment et, leading to states s' at time $t + 1$, and $R(s, s')$, which is the potential reward value after the transition from states s to s' as a result of action a.

A. Planning strategy

The primary goal of this part is to find a reasonable route from where you are now to where you want to go [21]. Equation (2) is the abbreviation for the planning strategy:

$$\Gamma_{t:t+T} = f_{\text{plan}}(l_{0:t}, g, mt) \quad (2)$$

Assuming that the path that is being planned is represented by the symbol $\Gamma_{t:t+T}$, the planning function is indicated by the symbol f_{plan} , the locations of mobile robot from time 0 to time t are represented by the symbol $l_{0:t}$, the desired location is represented by the symbol g , and the map that is formed in the environment at time t is known as mt .

B. Mapping strategy

The mapping segment, also known as where SLAM stands for simultaneous localization and mapping, analyzes the sensor data methodically to provide tracking the location and surroundings using mobile robot in real-time [21]. Here is the expression for the SLAM function:

$$mt, lt = f_{\text{map}}(lt-1, mt-1, ct, ot) \quad (3)$$

In the context within the contexts of oversight and management represented as ct and ot respectively, f_{map} denotes the process of mapping. The position of the mobile robot (MR) is indicated by $lt-1$ at time $t-1$, and $mt-1$ refers to the map generated across the environment at time $t-1$.

C. Mobile robot learning

Mobile robot (MR) navigation is a common problem in robotics for multispectral applications in various sectors. RL is widely used in many different areas, such as healthcare, robotics, smart surveillance systems, HCLs, and many more. We just tackle the problem of indoor navigation rather than trying to do a thorough analysis for an area this large. One common objective of navigation is to reach a destination as quickly as possible without crashing into anything. The capacity to understand its location is vital for an autonomous MR, as it determines the robot's next move based on its current location. The availability of an environment is maintained in different mapping-based robot navigation systems since the environment was first described in a metric map. In this case, MR picks up on its environment via the RL algorithm's use of DQN and DDQN methods. It is possible for MR to acquire knowledge about its surroundings without prior mapping thanks to RL algorithms. The majority of RL-based methods employed by robots today are for navigational tasks that include correlating the management approach with unprocessed inputs from sensors.

D. Deep Q-networks

For the purpose of determining the value of a particular action under every circumstance, a model-free reinforcement learning approach known as qualitative learning is used. Applying a DCNN with layers of convolutional filtering, visual DeepMind method mimics the effects underlying fields of reception. Researchers, Mnih et al. [18] found that after 49 games, the network that had been trained could reach a human-level performance, in a 2015 study released by Nature. DQN utilizes Q-learning, which whereby DCNN

stands out as the actionable value equation while the instruction of its neurons is done by the monetary input given by the player's actions. A gradient descent technique is employed to update the settings of the neural network as needed [30]. The function of loss of the DQN is shown in Equation (4):

$$L(\theta_i) = F \left[\left(R + \lambda \max_{A'} Q(S', A'; \theta_i^-) - Q(S, A; \theta_i) \right)^2 \right] \quad (4)$$

The discount component implies represented by $\lambda \in [0, 1]$, S denotes the condition, R signifies the recompense, A represents the action, $S' = S_{t+1}$, $A' = A_{t+1}$, and θ_i represents a value associated with the existing Q-network $Q(S, A; \theta_i)$ which is duplicated to θ_i^- for the target Q network $Q(S', A'; \theta_i^-)$ in every stage of n -time.

E. Mobile robot learning

DQN helped spread DRL, but it has drawbacks including an inflated action value function estimate that delays the process. After realizing that the value function is overestimated whenever the DQN calculates the temporal-difference (TD) error and the same Q network is utilized for both action selection and function that determines worth computation, Van Hasselt et al. proposed the DDQN method [31]. With DDQN's goal Q network estimating the stated optimum taking part in it and the present Q network selecting it, it makes advantage of a dual network design for the desired Q-function. To reduce the possibility of overestimation, two sets of variables are used: action decision and evaluation of policy actions [32]. As shown in Eq. (5), t . The loss equation of the DDQN as:

$$L(\theta_i) = F \left[\left(R + \lambda Q(S', \arg \max_{A'} Q)(S', A'; \theta_i; \theta_i^-) - Q(S, A; \theta_i) \right)^2 \right] \quad (5)$$

III. SIMULATION AND EXPERIMENTAL ANALYSIS

In order to improve the capacity to generalize while minimizing the need for manually acquiring supervised information, researchers are focusing their attention on autonomous interaction with surrounds and self-learning approaches, particularly applicable to RL systems. The following is a description of the kind of navigational task that is meant to be presented in this article: As it navigates an unfamiliar, dispersed environment with obstacles like lengthy hallways and dead corners, a robot that moves with dispersed local distance measurement sensors provides all the tools it needs for reaching a regular target. Due to the fact that it initially has no information about its surroundings, the MR is able to construct an independent environmental model via the process of inquiry. In addition to the dynamic and unexpected character of the surroundings, the presence of dynamic impediments contributes greatly to the complexity of the environment. Within dynamic, previously uncharted regions, we put into action a regional plan that is based on RL and converts visible sensory input into activities for the purpose of ensuring safe navigation and avoiding impediments. To facilitate navigation in complex scenarios and to enhance performance, we include a number of regular patterns into the fundamental RL techniques. Due to the fact that the memory

section is able to consider many sensor perceptions rather than relying just on the current perception, we predict that the navigation performance will be improved in more complex scenarios, such as alleys, lengthy halls, and concave areas. The simulation is intended to showcase the beginning and ending locations for the navigation of MR. Figure 2 depicts a binary occupancy grid map that contains various kinds of obstacles as well as the start and target positions for robot route planning. This map is made available for usage in the simulation.

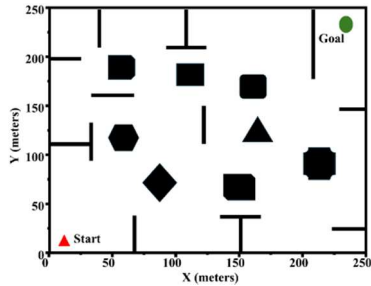


Fig. 2. Navigational binary saturation grids mapping

Within the framework of the standard robot navigation approach, the capacity for autonomous learning is not there. MRs are required to have the ability to autonomously explore unfamiliar locations and identify eliminate static and dynamic obstacles. A robot can navigate by making decisions based on its unique skills and sensor inputs to reach its desired destination quickly and effectively. Navigation provides incomplete comprehension of the robot's surrounds as well as a desired place or group of locations. Throughout the instruction stage for the neural network, it is possible to expand the process into a data set of available training scenarios. This allows for the procurement of plausible replies for analogous scenarios that may not have been seen in demonstrations. The suggested method is put through simulation testing in order to evaluate its effectiveness. When it comes to environmental mapping, the proportions are 250 meters by 250 meters. There is no particular pattern to the placement of the obstacles across the environment. On the map, the beginning point coordinate of the robot is shown as a red triangle, which is situated at the coordinates (10, 10). The destination position, on the other hand, is depicted as a green circle, which is situated at the intersection of (240, 240). In order to accomplish its mission, the robot must first start at the beginning point and then choose the most efficient path to travel in order to get to the destination without encountering any obstacles. This experiment makes use of a number of hyperparameters, and Table 1 provides an illustration of the list of hyperparameters and their explanations.

TABLE I. THE HYPERPARAMETERS ARE LISTED

System configuration	
Python version	2.9.2
TensorFlow version	2.9.2
NumPy rate	1.24.4
Learning rate	0.001
RAM	16GB
Activation function	ReLu
Gym version	0.26.2

During the process of training an MR, it is necessary to subject it to a number of unforeseen situations, which are known to as episodes. Every episode has a specific objective

that it aims to accomplish. Moreover, the number of challenges that are presented in each episode is different. There are some of these impediments that are believed to be static, while there are others that are thought to be dynamic barriers. When the robot is trained with a greater number of episodes, it will grow more adept at navigating the changing environment in which it is placed. For the purpose of functioning in simulation settings, the technique that has been described has been devised. Due to the fact that these paths are not the most efficient routes, we are investigating the ways in which they vary from one another. These two distinct habitats for the navigation of mobile robots are shown in Figure 3. Both of these environments have the beginning and objective locations predetermined.



Fig. 3. Different environments

After that, the result is compared with the most current reading of the position of the barriers that are constantly altering. Both of these readings are taken into consideration. After that, the information is sent via the array of Current Obstacles, where each value is assigned to a node inside the array that corresponds to it respectively. There is a certain event that the influence map has to remember for a specific period of time, and the Momentum parameter, which is responsible for the speed of changes, is the one that determines the pace of changes. The value of this coefficient will alter in accordance with the pace of change that is taking place in the real environment according to the circumstances. This is a depiction of the navigation map that was built in the Rviz software for the different ambient settings. Figure 4 illustrates that representation.

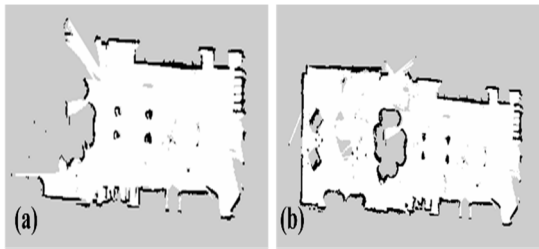


Fig. 4. A SLAM algorithm that takes into account the various ambient surroundings

In the beginning, the Adam optimizer method was the sole one that was used in the simulation that was carried out for experimental research. using the simulation environment in order to analyze the three algorithms based on their distinct rates of performance. It is vital to do this in order to offer a consistent quantitative evaluation of the approach and to show that limited state-space analysis methods may work in robots. In order to verify the suggested method for a total of one hundred episodes, we contrasted this strategy by the utilization of three separate optimizers, namely the Adam optimization method and the RMSprop optimizer method. Figure 5 depicts

an analysis with the incentive value obtained from the Adam optimizer and the RMSprop optimizer according to their respective algorithms, which is an example of comparison. During the simulation, the Adam optimizer is the one that obtains the biggest reward, as seen in Figure 5.

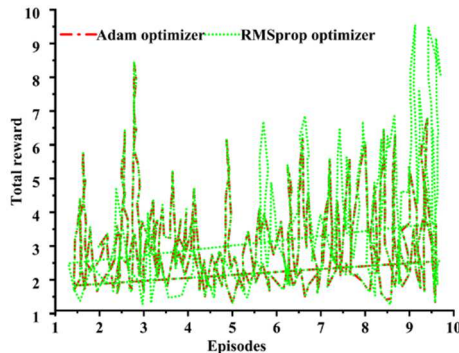


Fig. 5. Comparison graphs of incentive value created with Adam and RMSprop optimizers

The RL approach has been created in order to verify its superior sampling performance, fast mobility, large target-reach success rate, and increased extrapolation with respect to a variety of motion scenarios and surroundings. This was accomplished by doing relevant research and comparing the results with other optimization algorithms. Within the context of evaluating the efficacy of our technique, the suggested method has contrasted with PPO. A method known as PPO is a kind of reinforcement learning that the computational robot and automaton enabling do difficult projects via training. A comparison of the suggested approach method and the PPO method and their respective reward value evaluations is shown in Figure 6. In the comparing graph, it is clear that the Q-learning algorithm method performs better than the PPO algorithm.

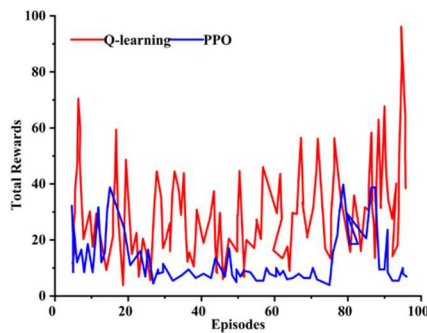


Fig. 6. A contrast between the PPO algorithm's payout and that of the indicated algorithm

IV. CONCLUSION

There are a number of benefits to using the suggested method instead of the aforementioned alternatives. One advantage is that robot dynamics and the surrounding environment don't have to be explicitly modeled. Secondly, it requires no human intervention to adjust to new settings or tasks; all control parameters are automatically fine-tuned. As a third benefit, it may become better with use of past mistakes and other examples. The accuracy rate of this method is greater than that of the conventional MR control technique. However, indoor plains are where most of these DRL-based movement control systems really shine. Only a small number of studies have managed to keep MR under control in a completely foreign setting, as far as we are aware. In this

study, we provide a DRL-based end-to-end mobility control method to help MR overcome the limitations of the previously mentioned methodologies and function more efficiently in complex and unfamiliar environments. It is possible that the real-world application situation will differ from the one simulated by the suggested technique. Our main focus here is on the tracking problem that non-holonomic robots face while operating in environments where their trajectory is always under our control. Presenting a DRL approach with outstanding training and adaptability capability across various settings and robotic systems, while also being able to swiftly and securely traverse complex environments, is our aim.

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REFERENCES

- [1] Y. Gao, S. Liu, and L. Yang, "Artificial intelligence and innovation capability: A dynamic capabilities perspective," *International Review of Economics & Finance*, vol. 98, p. 103923, Mar. 2025, doi: 10.1016/j.iref.2025.103923.
- [2] M. Soori, B. Arezoo, and R. Dastres, "Artificial intelligence, machine learning and deep learning in advanced robotics, a review," *Cognitive Robotics*, vol. 3, pp. 54–70, 2023, doi: 10.1016/j.cogr.2023.04.001.
- [3] S. Rokhsaritalemi, A. Sadeghi-Niaraki, and S.-M. Choi, "A Review on Mixed Reality: Current Trends, Challenges and Prospects," *Applied Sciences*, vol. 10, no. 2, p. 636, Jan. 2020, doi: 10.3390/app10020636.
- [4] R. Raj and A. Kos, "Intelligent mobile robot navigation in unknown and complex environment using reinforcement learning technique," *Sci Rep*, vol. 14, no. 1, p. 22852, Oct. 2024, doi: 10.1038/s41598-024-72857-3.
- [5] D. Debnath, F. Vanegas, J. Sandino, A. F. Hawary, and F. Gonzalez, "A Review of UAV Path-Planning Algorithms and Obstacle Avoidance Methods for Remote Sensing Applications," *Remote Sensing*, vol. 16, no. 21, p. 4019, Oct. 2024, doi: 10.3390/rs16214019.
- [6] J. Rascon Enriquez, B. Castillo-Toledo, S. Di Gennaro, and L. A. García-Delgado, "An algorithm for dynamic obstacle avoidance applied to UAVs," *Robotics and Autonomous Systems*, vol. 186, p. 104907, Apr. 2025, doi: 10.1016/j.robot.2024.104907.
- [7] D. Debnath, F. Vanegas, S. Boiteau, and F. Gonzalez, "An Integrated Geometric Obstacle Avoidance and Genetic Algorithm TSP Model for UAV Path Planning," *Drones*, vol. 8, no. 7, p. 302, Jul. 2024, doi: 10.3390/drones8070302.
- [8] K. Ofosu-Ampong, "Artificial intelligence research: A review on dominant themes, methods, frameworks and future research directions," *Telematics and Informatics Reports*, vol. 14, p. 100127, Jun. 2024, doi: 10.1016/j.teler.2024.100127.

- [9] F. Rubio, F. Valero, and C. Llopis-Albert, "A review of mobile robots: Concepts, methods, theoretical framework, and applications," *International Journal of Advanced Robotic Systems*, vol. 16, no. 2, p. 1729881419839596, Mar. 2019, doi: 10.1177/1729881419839596.
- [10] R. Raj and A. Kos, "A Comprehensive Study of Mobile Robot: History, Developments, Applications, and Future Research Perspectives," *Applied Sciences*, vol. 12, no. 14, p. 6951, Jul. 2022, doi: 10.3390/app12146951.
- [11] Z. Zhou et al., "A safe reinforcement learning approach for autonomous navigation of mobile robots in dynamic environments," *CAAI Trans on Intel Tech*, p. cit2.12269, Oct. 2023, doi: 10.1049/cit2.12269.
- [12] K. Katona, H. A. Neamah, and P. Korondi, "Obstacle Avoidance and Path Planning Methods for Autonomous Navigation of Mobile Robot," *Sensors*, vol. 24, no. 11, p. 3573, Jun. 2024, doi: 10.3390/s24113573.
- [13] R. Alharthi, I. Noreen, A. Khan, T. Aljrees, Z. Riaz, and N. Innab, "Novel deep reinforcement learning based collision avoidance approach for path planning of robots in unknown environment," *PLoS ONE*, vol. 20, no. 1, p. e0312559, Jan. 2025, doi: 10.1371/journal.pone.0312559.
- [14] W. Xia, Y. Lu, W. Xu, and X. Xu, "Deep reinforcement learning based proactive dynamic obstacle avoidance for safe human-robot collaboration," *Manufacturing Letters*, vol. 41, pp. 1246–1256, Oct. 2024, doi: 10.1016/j.mfglet.2024.09.151.
- [15] F. Kessler, J. Frankenstein, and C. A. Rothkopf, "Human navigation strategies and their errors result from dynamic interactions of spatial uncertainties," *Nat Commun*, vol. 15, no. 1, p. 5677, Jul. 2024, doi: 10.1038/s41467-024-49722-y.
- [16] Y. Cao and R. B. Gopaluni, "Deep Neural Network Approximation of Nonlinear Model Predictive Control," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 11319–11324, 2020, doi: 10.1016/j.ifacol.2020.12.538.
- [17] D. Arce, J. Solano, and C. Beltrán, "A Comparison Study between Traditional and Deep-Reinforcement-Learning-Based Algorithms for Indoor Autonomous Navigation in Dynamic Scenarios," *Sensors*, vol. 23, no. 24, p. 9672, Dec. 2023, doi: 10.3390/s23249672.
- [18] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, Feb. 2015, doi: 10.1038/nature14236.
- [19] Y. Pan, H. Jiang, H. Yang, and J. Zhang, "A Novel Method for Improving the Training Efficiency of Deep Multi-Agent Reinforcement Learning," *IEEE Access*, vol. 7, pp. 137992–137999, 2019, doi: 10.1109/ACCESS.2019.2942635.
- [20] J. Chung, J. Fayyad, Y. A. Younes, and H. Najjaran, "Learning team-based navigation: a review of deep reinforcement learning techniques for multi-agent pathfinding," *Artif Intell Rev*, vol. 57, no. 2, p. 41, Feb. 2024, doi: 10.1007/s10462-023-10670-6.
- [21] H. Li, Q. Zhang, and D. Zhao, "Deep Reinforcement Learning-Based Automatic Exploration for Navigation in Unknown Environment," *IEEE Trans. Neural Netw. Learning Syst.*, vol. 31, no. 6, pp. 2064–2076, Jun. 2020, doi: 10.1109/TNNLS.2019.2927869.
- [22] M. Cai, E. Aasi, C. Belta, and C.-I. Vasile, "Overcoming Exploration: Deep Reinforcement Learning for Continuous Control in Cluttered Environments from Temporal Logic Specifications," *IEEE Robot. Autom. Lett.*, vol. 8, no. 4, pp. 2158–2165, Apr. 2023, doi: 10.1109/LRA.2023.3246844.
- [23] W. Yan, J. Sun, Z. Li, and G. Wang, "Decentralized Multi-Robot Navigation in Unknown Environments via Hierarchical Deep Reinforcement Learning," in *2023 42nd Chinese Control Conference (CCC)*, Tianjin, China: IEEE, Jul. 2023, pp. 4292–4297. doi: 10.23919/CCC58697.2023.10240139.
- [24] Y. Xue and W. Chen, "A UAV Navigation Approach Based on Deep Reinforcement Learning in Large Cluttered 3D Environments," *IEEE Trans. Veh. Technol.*, vol. 72, no. 3, pp. 3001–3014, Mar. 2023, doi: 10.1109/TVT.2022.3218855.
- [25] J. Wang and R. Huang, "A Mapless Navigation Method Based on Deep Reinforcement Learning and Path Planning," in *2022 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Jinghong, China: IEEE, Dec. 2022, pp. 1781–1786. doi: 10.1109/ROBIO55434.2022.10011923.
- [26] J. Zeng, R. Ju, L. Qin, Y. Hu, Q. Yin, and C. Hu, "Navigation in Unknown Dynamic Environments Based on Deep Reinforcement Learning," *Sensors*, vol. 19, no. 18, p. 3837, Sep. 2019, doi: 10.3390/s19183837.
- [27] W. Zhu and M. Hayashibe, "A Hierarchical Deep Reinforcement Learning Framework With High Efficiency and Generalization for Fast and Safe Navigation," *IEEE Trans. Ind. Electron.*, vol. 70, no. 5, pp. 4962–4971, May 2023, doi: 10.1109/TIE.2022.3190850.
- [28] H. Haile, K.-J. Grinnemo, S. Ferlin, P. Hurtig, and A. Brunstrom, "End-to-end congestion control approaches for high throughput and low delay in 4G/5G cellular networks," *Computer Networks*, vol. 186, p. 107692, Feb. 2021, doi: 10.1016/j.comnet.2020.107692.
- [29] S. Fianu and L. B. Davis, "Heuristic algorithm for nested Markov decision process: Solution quality and computational complexity," *Computers & Operations Research*, vol. 159, p. 106297, Nov. 2023, doi: 10.1016/j.cor.2023.106297.
- [30] K. Zhu and T. Zhang, "Deep reinforcement learning based mobile robot navigation: A review," *Tsinghua Sci. Technol.*, vol. 26, no. 5, pp. 674–691, Oct. 2021, doi: 10.26599/TST.2021.9010012.
- [31] H. Van Hasselt, A. Guez, and D. Silver, "Deep Reinforcement Learning with Double Q-Learning," *AAAI*, vol. 30, no. 1, Mar. 2016, doi: 10.1609/aaai.v30i1.10295.
- [32] Y. Yin, L. Zhang, X. Shi, Y. Wang, J. Peng, and J. Zou, "Improved Double Deep Q Network Algorithm Based on Average Q-Value Estimation and Reward Redistribution for Robot Path Planning," *CMC*, vol. 81, no. 2, pp. 2769–2790, 2024, doi: 10.32604/cmc.2024.056791.-0.